Chapter 24
Credit Risk and Default Modeling

Chapter 9 computed a credit valuation adjustment and took two inputs as given: a default rate read off a CDS spread through the credit triangle, assumed flat, and a recovery rate assumed to be 40%. It pointed here for the machinery behind both. This chapter supplies it, and then asks what the inputs actually mean.

Credit risk is among the oldest financial risks and the largest one most banks carry. Bank loan books, the corporate and sovereign bond markets, the credit default swaps of Chapter 25, the securitizations of Chapter 26, and the counterparty exposures of Chapter 27 all price off the two quantities this chapter is about, and anyone who lends, holds a bond, or faces a counterparty is exposed to them whether or not they model them. Section 24.16 says who those holders are and how the risk has moved between them since the reforms.

The subject divides cleanly. Credit risk is two questions (whether a borrower defaults, and how much is recovered if it does), and almost every difficulty in this part of the book comes from market prices identifying only their product. Sections 24.1 to 24.4 establish that, with the credit triangle and the hazard-curve bootstrap on the way. Sections 24.5 to 24.8 are the two families of model that try to say more: structural models, which explain default as an option finishing out of the money and fail at short maturities, and reduced-form models, which assume it and fit anything. Sections 24.9 to 24.12 are the gap between what prices imply and what happens: risk-neutral against historical default rates, ratings and migration, and the sovereign case in which default is a choice. Section 24.13 introduces the correlation machinery, Section 24.14 turns it into the credit VaR against which a loan book is capitalized, Section 24.15 reads a hazard rate from a bond price and sets it against the CDS, and Section 24.17 records where the models came from.

24.1 THE TWO QUESTIONS

A defaultable bond promises \$100 and delivers either \$100 or something less. Valuing it requires a probability and a payoff.

Default probability. The default probability is not a single number but a curve: the chance of default by each future date. Write τ for the random time of default and $Q(t) = \mathbb{Q}(\tau > t)$ for the **survival probability**, the chance that the borrower is still performing at t. The convention is to work not with Q itself but with the **hazard rate** $\lambda(t)$, the probability of defaulting in the next instant given survival to t, per unit of time:

$$\lambda(t) = \lim_{\Delta t \to 0} \frac{\mathbb{Q}(t < \tau \le t + \Delta t \mid \tau > t)}{\Delta t} = -\frac{Q'(t)}{Q(t)} = -\frac{d}{dt} \ln Q(t) \tag{24.1}$$

The second equality is the definition of a conditional probability: the probability of defaulting in $(t, t + \Delta t]$ is $Q(t) - Q(t + \Delta t)$, and dividing by the probability $Q(t)$ of being there to default gives the conditional rate. Integrating the last form gives the survival probability to time T,

$$Q(T) = \exp\left(- \int_0^T \lambda(u)\, du \right) \tag{24.2}$$

For a constant hazard this is $e^{-\lambda T}$, and the density of default at t is $\lambda e^{-\lambda t}$.

The hazard rate is not the default probability. A 3% hazard rate is an intensity of 3% per year: *given survival to the start of a year*, the chance of defaulting during it is $1 - e^{-0.03} = 2.96\%$, and over ten years the cumulative default probability is $1 - e^{-0.3} = 25.9\%$, not 30%. The distinction matters as soon as the horizon is long or the borrower is weak.

Under which probability? Two versions of every quantity above will appear in this chapter, and they are not close. The one that prices a contract is the risk-neutral probability of Chapters 13 and 28 (written $\mathbb{Q}$ there and here), and it is the one the market reveals: a CDS spread read through the next two sections gives a risk-neutral λ, because the spread is a price. The one that forecasts is the **real-world** probability ($\mathbb{P}$), estimated from default records. Everything up to Section 24.8 is risk-neutral, and should be read as pricing rather than prediction; Section 24.9 sets the two side by side and finds them apart by a factor of several.

Recovery. Recovery is what the claim is worth after default, conventionally quoted as a fraction R of face value, with $1 - R$ the **loss given default**. Forty percent is the market's standard assumption for senior unsecured corporate debt, and Section 24.4 explains why that number is used far beyond the evidence supporting it. Everything else in this chapter is a way of estimating one of these two, or an admission that only their product is observable.

24.2 THE CREDIT TRIANGLE

A credit default swap pays $1 - R$ per unit of notional if the **reference entity** (the borrower the contract is written on) defaults before maturity T, in exchange for a spread s per year paid by the protection buyer until default or maturity. Its par spread is the s at which the two legs have the same value today, and both legs are integrals over the risk-neutral survival curve of Section 24.1. Write $P(0,t)$ for the risk-free discount factor and take the premium to be paid continuously for the moment. The premium leg collects $s\,dt$ over each instant the borrower is still alive, so its value is

$$\text{premium leg} = s \int_0^T P(0,t)\, Q(t)\, dt.$$

The protection leg pays $1 - R$ at the moment of default, and by (24.1) the probability of that moment falling in $(t, t + dt]$ is $\lambda(t)Q(t)\,dt$, the hazard rate times the probability of having survived to t, so

$$\text{protection leg} = (1 - R) \int_0^T P(0,t)\, \lambda(t)\, Q(t)\, dt.$$

If the hazard rate and the interest rate are both flat, $Q(t) = e^{-\lambda t}$ and $P(0,t) = e^{-rt}$, the integrand decays at $r + \lambda$, and

$$\text{protection leg} = (1 - R)\,\lambda \int_0^T e^{-(r+\lambda)t}\, dt = (1 - R)\,\lambda\, \frac{1 - e^{-(r+\lambda)T}}{r + \lambda}.$$

For a flat hazard rate on any discount curve, λ comes outside the integral, the same integral $\int_0^T P(0,t)Q(t)\,dt$ appears in both legs, and setting them equal cancels it:

$$\boxed{s = \lambda\,(1 - R)} \tag{24.3}$$

Neither the discount curve nor the level of the survival probability remains in the answer; the par spread is the default rate times the loss when default happens, which is what the price of insurance ought to be. The result is usually called the **credit triangle**, because any two of s, λ and $1 - R$ give the third. Reading a spread as a hazard rate therefore requires only a recovery assumption, not a discount curve, which is why traders can do the conversion mentally; and a change in the discount curve alone barely moves the par spread, which is why CDS

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spreads did not move mechanically when the discounting convention changed in the reforms of Chapter 4. The triangle is usually presented as an approximation valid for small λ, and the standard statement puts the error in the wrong place.

Example 24.1 The credit triangle is exact for a continuous premium, and the error people quote is the accrued premium

Price a five-year CDS with a flat hazard rate $\lambda = 1\%$, discounting at $r = 3\%$, recovery $R = 40\%$, three ways: with the premium paid continuously; quarterly, with the premium accrued since the last payment date paid on default, as real contracts do; and quarterly with no accrual. The protection leg is the same in all three; from the flat-rate formula above, with $r + \lambda = 0.04$, it is $0.6 \times 0.01 \times (1 - e^{-0.2})/0.04 = 0.027190$ per unit of notional. The premium legs are the spread s times an annuity, and the annuities differ: 4.5317, 4.5148 and 4.5091 ([TN25] works them out one quarter at a time). The par spread sets each premium leg equal to the protection leg:

Continuous premium	$4.5317\, s = 0.027190$	$\Rightarrow$	$s = 60.00$ bp
Quarterly, with accrual	$4.5148\, s = 0.027190$	$\Rightarrow$	$s = 60.23$ bp
Quarterly, no accrual	$4.5091\, s = 0.027190$	$\Rightarrow$	$s = 60.30$ bp

The first is $\lambda(1 - R) = 0.01 \times 0.6$ exactly, as the derivation of (24.3) says it must be. The second is 0.376% higher, because a premium paid at the end of a quarter is discounted for half a quarter longer, on average, than the same money paid continuously through it; to first order the excess is $r\Delta/2$ for a payment interval $\Delta = 0.25$, and Chapter 25 carries the same effect as about a basis point on a 250 bp contract. The third drops the accrued premium, and that is the "approximation error" of the textbook statement: a tenth of a percent here. It is the value of a contract feature, not an error in the triangle, and unlike the 0.376% it grows with the hazard rate: at $\lambda = 20\%$ the three spreads are 1200.00, 1204.47 and 1235.17 bp, and the missing accrual is worth 2.5%.

24.3 BOOTSTRAPPING A HAZARD CURVE

Spreads are quoted at several maturities and they are not flat. Extracting the implied hazard curve is the credit analog of Chapter 4's zero-curve bootstrap: work outward, taking each quote as determining the hazard rate over the interval that quote is the first to cover.

Example 24.2 From a spread curve to a hazard curve

An issuer's CDS curve is quoted at the maturities below; recovery is assumed at 40% and rates at 3%:

Maturity	Spread (bp)	Bootstrapped λ	Credit-triangle guess
1y	80	1.333% (0–1y)	1.333%
3y	130	2.617% (1–3y)	2.167%
5y	170	3.998% (3–5y)	2.833%
7y	190	4.222% (5–7y)	3.167%
10y	200	3.885% (7–10y)	3.333%

The second row shows the step. The one-year quote fixes $\lambda_1 = 0.0080/0.6 = 1.333\%$ exactly by (24.3), because a single bucket is the constant-hazard case. The three-year quote must then be reproduced by a contract whose first year runs at λ_1 and whose next two years run at an unknown λ_2. With a continuously paid premium both legs are integrals of $e^{-rt}Q(t)$, and the integral splits at the knot: write $A_1 = \int_0^1 e^{-(r+\lambda_1)t}\,dt = 0.9786$ for the first year and $A_2 = \int_1^3 e^{-rt}Q(t)\,dt = Q(1)\,e^{-r}\int_0^2 e^{-(r+\lambda_2)u}\,du$ for the next two (on $[1,3]$, $Q(t) = Q(1)\,e^{-\lambda_2(t-1)}$; substitute $u = t - 1$), where $Q(1) = e^{-\lambda_1} = 0.9868$ is the probability of reaching the second bucket alive. The par

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condition is

$$s_3\,(A_1 + A_2) = (1 - R)\,(\lambda_1 A_1 + \lambda_2 A_2),$$

one equation in λ_2, since A_2 depends on it. At $\lambda_2 = 2.617\%$, $A_2 = 1.8115$; the premium leg is $0.0130 \times 2.7902 = 0.036272$ and the protection leg is $0.6 \times (0.013049 + 0.047405) = 0.036272$, so the equation balances. Each later bucket repeats the step with one more known piece and one unknown. The credit triangle applied to the three-year spread gives $0.0130/0.6 = 2.167\%$ instead, close to the time-weighted average of the two bucket rates, $(1.333\% + 2 \times 2.617\%)/3 = 2.19\%$, but not the rate for years one to three.

The last column applies (24.3) directly to each quoted spread. It is the same as the bootstrapped rate for the first bucket and different after that, most of all in the 3–5 year bucket, and the reason is the reason a swap rate is not a forward rate: a spread is an average over the whole life of the contract, and the hazard rate for a later bucket must exceed that average when the curve is upward sloping.

At the 3–5 year bucket the difference is large: 3.998% against 2.833%, a factor of 1.41. Anyone using the five-year spread to price something that pays off only in years three to five will understate the hazard rate for those years by almost 30%.

The implied cumulative default probabilities are

	1y	3y	5y	7y	10y
Cumulative default probability	1.32%	6.36%	13.55%	20.55%	29.29%

Note the shape of the hazard curve: rising through seven years and then falling. An upward-sloping spread curve is the norm for **investment-grade** issuers, those rated BBB (Baa) or higher; an issuer rated below that line is **speculative grade**. The upward slope reflects the same mean reversion that Section 19.6 found in volatility: a currently healthy firm is more likely to deteriorate than to improve. The fall after seven years is arithmetic: the ten-year spread is only 10 bp above the seven-year one, so the last bucket cannot carry a hazard rate as high as the one before it. Distressed issuers show the same effect in extreme form, with an inverted spread curve, for the same reason inverted volatility curves appear in stress: if the firm survives the next year, the danger has largely passed and the hazard falls.

Figure 24.1 draws the table's last two columns.

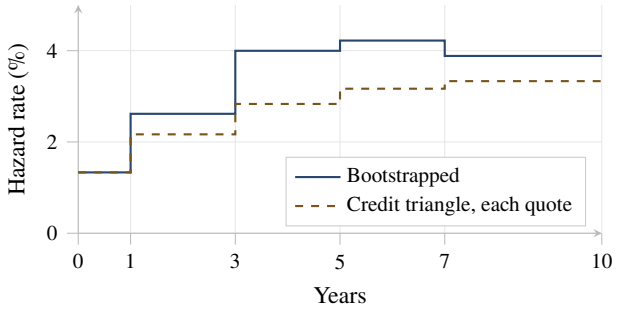


Figure 24.1: Hazard rates bootstrapped from the spread curve of Example 24.2, against the credit triangle applied to each quoted spread directly.

24.4 RECOVERY: THE OTHER UNKNOWN

The bootstrap above assumed $R = 40\%$. Where does that come from, and what happens if it is wrong? The answer to the first question is convention, supported by an average that conceals almost everything of interest.

Recovery varies by seniority. Table 24.1 gives Moody's long-run averages measured two ways: the market's bid for the defaulted debt shortly after default, close to what a CDS auction

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Table 24.1: Average recovery rates by priority of claim (%), issuer-weighted for trading prices. Moody's, March 2023 study.

	Trading price, 1983–2022	Ultimate, 1987–2022
First-lien bank loan	65.7	71.7
Second-lien bank loan	33.3	–
Senior unsecured bank loan	46.3	–
First-lien bond	55.0	61.2
Second-lien bond	44.2	–
Senior unsecured bond	37.7	47.1
Senior subordinated bond	31.0	–
Subordinated bond	32.0	27.8
Junior subordinated bond	22.3	–

sets, and the ultimate recovery when the default is resolved, usually a year or two later (North American nonfinancial issuers; the figures for all term loans and all senior secured bonds stand in the first-lien rows). The ranking broadly follows priority, from about two-thirds on first-lien loans to about a fifth on junior subordinated bonds, though not strictly. Senior unsecured bonds, the class the 40% convention describes, recovered 37.7% at the trading price and 47.1% at resolution; the two come from different samples, so the gap is not a measured increase.

Recovery is also correlated with the default rate. Recoveries are lowest in the years when defaults are highest (Altman, Brady, Resti, and Sironi 2005), because a firm defaulting in a recession is selling its assets into a market full of other distressed sellers.

Example 24.3 Hazard and recovery are not separately identified

A five-year CDS trades at 170 bp. Which (λ, R) pair produced it?

Recovery	Implied λ	5-year default probability
20%	2.125%	10.08%
40%	2.833%	13.21%
60%	4.250%	19.14%
80%	8.500%	34.62%

All four price the CDS identically. The market observes only the product $\lambda(1 - R)$, so the recovery assumption is not extracted from the spread—it is imposed on it.

Usually this does not matter, because the thing being priced also depends only on the product. When it does not, the split matters a great deal, and three cases show how much.

A digital default swap pays a fixed amount on default regardless of recovery, so its value depends on λ alone. The table's four rows price it roughly in proportion to the last column, a range of more than three to one; between the 40% and 60% rows alone a five-year digital is worth 45% more. The contract has only ever traded thinly, mostly between dealers separating a view on default from a view on recovery; its use here is as the sharpest test of the split.

A first-to-default basket pays $1 - R$ on the first default among a group of borrowers. To first order its value is the sum of the borrowers' $\lambda_i(1 - R)$, the product again, so most of the ambiguity cancels; what survives comes from the probability of at least one default being concave in the λ_i. On five independent borrowers each quoted at 170 bp, the basket prices about 14% lower at 60% recovery than at 40%, where the digital prices 45% higher.

And any calculation of expected loss for capital or provisioning needs the split, because regulation asks for probability of default and loss given default separately.

The practical convention is to fix R by seniority and sector and let λ absorb everything, which is defensible for pricing and indefensible for risk measurement, for the reasons the case study below sets out.

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Case Study 24.1 Why the 40% convention survives evidence against it

The recovery assumption of Section 24.4 is used with a confidence the data do not support, and the gap follows a pattern that recurs throughout this part of the book.

Realized recoveries are enormously dispersed. The 40% figure is a long-run average across senior unsecured corporate bonds. Year by year the average moves a good deal: senior unsecured bonds recovered 48.7% in 2021 and 31.1% in 2022, when defaults by Russian issuers (recovering 15.5%) and Chinese property developers (25.3%) pulled it down.

The dispersion is systematic, not noise. Recovery depends on seniority, on whether the debt is secured and by what, on the industry's asset tangibility, on the jurisdiction's insolvency regime, and on where in the credit cycle the default occurs. An asset-heavy firm in a creditor-friendly jurisdiction defaulting in a boom is a different proposition from an asset-light firm in a debtor-friendly one defaulting in a recession, and the two have almost nothing in common but the letter on the rating.

Recovery moves with the thing it is multiplied by. A portfolio model that fixes recovery at 40% and lets only the default rate vary understates tail losses twice over: once because more borrowers default, and once because each one recovers less. In a bad year in which twice the average number of borrowers default and recovery falls from 40% to 25%, the loss is $2 \times 0.75/0.60 = 2.5$ times the average year's, where the fixed-recovery model says 2.

So why does the convention hold? Because for pricing a CDS it does not matter. Example 24.3 showed that the spread identifies only $\lambda(1 - R)$, so a desk that fixes R and lets λ absorb the residual prices the contract correctly whatever R truly is. The convention is not a belief about recovery; it is a coordinate choice, and within its domain it is harmless.

The damage is done when the coordinate is exported. Multiplying the extracted λ back by the same 60% loss given default is harmless: it returns the spread, whatever the true recovery. The harm comes from pairing that λ with a recovery from somewhere else (a risk system that takes the λ extracted at 40% and applies its own loss-given-default estimate of 75% has used two recoveries for one borrower), or from feeding λ into a calculation that is nonlinear in the default probability, such as the worst-case default rate of Section 24.14, where the product no longer carries through. A capital calculation that requires probability of default and loss given default separately is asking for a decomposition the market never provided.

The general lesson is one this book has met before. Section 21.10 observed that a calibration encodes an assumption about what the model is for; Section 22.6 observed that VIX is well defined and is not the price of anything. A number can be exactly right for the purpose that produced it and badly wrong one step away, and credit is the area where that distance is shortest and least visible.

24.5 STRUCTURAL MODELS: MERTON

The two families of model differ in whether default is explained or assumed. Merton's insight is that a firm's equity is a call option on its assets. If the firm's assets are worth V_T at the date its debt of face value D matures, shareholders receive $\max(V_T - D, 0)$: they keep the surplus and walk away from the shortfall. The strike is the face value of the debt.

Figure 24.2 shows the consequence for the other side. Debt holders receive $\min(V_T, D)$, which is D less a put struck at D: lending is selling a put on the borrower's assets, and the credit spread is the premium on that put.

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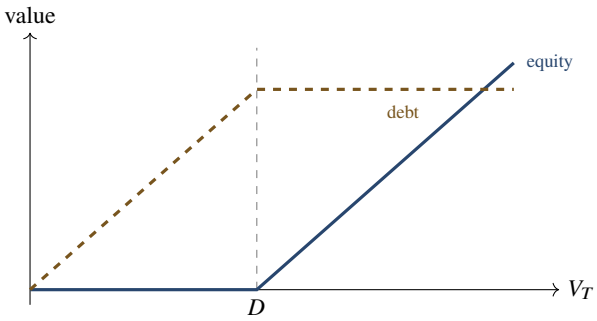


Figure 24.2: Equity and debt payoffs at the debt's maturity.

This is the sharpest available statement of the conflict Section 17.11 flagged. Equity is a call, so equity holders gain from volatility; debt is short a put, so debt holders lose from it. An executive paid in options is paid in a claim whose value rises with firm risk, and every increase in that risk, at a given firm value, is a transfer from creditors. Nothing in the payoff structure aligns the two, and no amount of governance language changes the derivative.

24.6 CALIBRATING MERTON FROM WHAT IS OBSERVABLE

Firm asset value and asset volatility are not observable. Equity value and equity volatility are, and two equations link them.

Write V for the firm's asset value today and σ_V for its volatility, with D and T the face value and maturity of the debt as in Section 24.5. The first equation is Chapter 15's formula applied to the call that equity is:

$$E = V\,N(d_1) - De^{-rT}N(d_2), \qquad d_1 = \frac{\ln(V/D) + \left(r + \tfrac{1}{2}\sigma_V^2\right)T}{\sigma_V\sqrt{T}}, \qquad d_2 = d_1 - \sigma_V\sqrt{T} \tag{24.4}$$

The second follows from applying Itô's lemma to E as a function of V, which is Chapter 14's calculation: equity volatility is asset volatility times the call's elasticity to the firm, $N(d_1)V/E$, the percentage change in equity value for a one percent change in asset value:

$$\sigma_E E = N(d_1)\,\sigma_V V \tag{24.5}$$

Two equations, two unknowns, and everything on the left-hand side is quoted. There is no closed form for V and σ_V; the pair is solved numerically, which is all Example 24.4 does.

The output the model is actually used for is the argument of the second normal:

$$\mathrm{DD} = d_2 = \frac{\ln(V/D) + \left(r - \tfrac{1}{2}\sigma_V^2\right)T}{\sigma_V\sqrt{T}} \tag{24.6}$$

This is the **distance to default** (how many standard deviations of asset value separate the firm from its debt), and the risk-neutral default probability is

$$\mathbb{Q}(V_T < D) = 1 - N(d_2) = N(-\mathrm{DD}).$$

Example 24.4 Distance to default from a share price

A firm has equity worth 40 with 55% volatility, and debt of face value 80 maturing in five years. Rates are 5%.

Solving the two equations simultaneously gives:

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Implied asset value	95.87
Implied asset volatility	27.2%
Distance to default	0.41 standard deviations
Risk-neutral 5-year default probability	34.26%
Implied credit spread	218 bp

The last row follows from the first. Debt is the firm less its equity, $95.87 - 40 = 55.87$, against $80e^{-0.25} = 62.30$ if it were riskless, and the continuously compounded spread that discounts one to the other is

$$s = -\frac{1}{T}\ln\frac{55.87}{62.30} = 2.18\%.$$

The first two rows can be checked by substitution. At the solution, before rounding, $d_1 = 1.0131$ and $d_2 = 0.4054$, so (24.4) gives $95.87 \times 0.8445 - 62.30 \times 0.6574 = 80.96 - 40.96 = 40.00$, the observed equity value, and (24.5) gives $\sigma_E = 0.8445 \times 0.272 \times 95.87/40 \approx 55\%$, the observed volatility. The same $d_2 = 0.4054$ is the third row, and $N(-0.4054) = 34.26\%$ is the fourth.

The distance to default of (24.6) is the model's most durable output. What Merton's model establishes is a monotonic relationship between distance to default and default probability: a firm further from its **default point** (the asset value at which it defaults, the face value D in Merton's model), measured in standard deviations of its own asset value, defaults less often. That ordering is more robust than the default probability that follows from it, because it does not require the normality assumption to be right, only the monotonicity, which is the license the commercial mapping below takes. Two observations about the numbers follow.

Equity is twice as volatile as the firm. The ratio of 55% to 27.2% is the equity's elasticity to the firm, $N(d_1)V/E = 0.8445 \times 95.87/40 = 2.02$: the leverage $V/E = 2.40$ scaled down by the call's delta. It rises as the firm approaches default, which is why distressed equities are so volatile and why Section 19.9's leverage explanation of the skew is this model in miniature.

And the implied recovery is high. The expected loss is 10.3% of the riskless value ($1 - 55.87/62.30$) against a 34.26% default probability, an implied recovery of $1 - 10.3/34.26$, near 70%. Merton recovers too much because, on default, the bondholders take over the whole of the firm's assets at no cost. With 27% asset volatility those assets are still worth on average about 70% of the debt's face when the firm defaults (55.88 of 80, a risk-neutral expectation, which is exactly the implied recovery), and the model has no bankruptcy costs or fire-sale discounts to reduce that.

What the commercial versions change. The framework became a product before it became a consensus, and the standard implementation is the one built by **KMV** (the firm Kealhofer, McQuown and Vasicek founded, named from their initials), now part of Moody's Analytics. It departs from Merton in three ways, each replacing one of the model's weakest assumptions with data rather than with more theory.

The default point is not the face value of debt. Firms do not fail the moment assets touch total liabilities, because long-dated debt need not be repaid yet. The default point is set at short-term debt plus half of long-term debt. That rule is not derived from the model; KMV chose it because, in its database of defaulted firms, asset value at the time of default was typically around that level.

The default probability does not come from the normal distribution. Instead of reading $N(-\text{DD})$, distance to default is mapped to an **expected default frequency** (EDF) through an empirical table built from a large default history, and the distance itself is computed with the firm's expected asset return in place of r, because the target is a real-world frequency rather than a price. This is the change that matters, because Section 24.7 is about to show that the normal tail is far too thin at short horizons. It is also why distance to default outlives the model that produced it: the mapping uses only the ordering, so (24.6) can be right about which firms are riskier while $N(-\text{DD})$ is wrong about how risky they are.

And the calibration is iterative. Rather than solving (24.4) and (24.5) once, the implied

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asset series is used to re-estimate σ_V and the system is re-solved to convergence, which is more stable when the equity volatility going in is itself an estimate.

The pattern recurs. A commercial version of an academic model is usually that model with its weakest assumption replaced by a fitted table, and the table holds both the value and the secrecy. Chapter 26 shows the same substitution made for default correlation, and made badly.

24.7 WHAT MERTON GETS WRONG

The model's characteristic failure is at the short end, and it can be measured precisely.

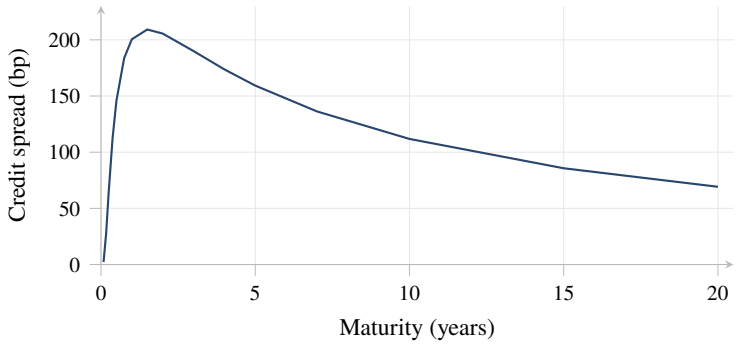


Figure 24.3: Credit spread by maturity in Merton's model. Assets 100, debt 80, asset volatility 25%, $r = 5\%$.

Figure 24.3 has the right shape in the middle and the wrong shape at the left edge.

Maturity	Spread	Default probability
1 week	0.0000 bp	6×10^{-9}%
1 month	2.2 bp	0.09%
3 months	64.6 bp	3.42%
1 year	200.5 bp	16.66%
2 years	205.6 bp	23.05%

The spread goes to zero as maturity shortens. A firm whose assets exceed its debt cannot default tomorrow if asset value follows a continuous path, because it would have to pass through every intermediate level first and there is no time for it to travel.

Real short-dated credit does not behave this way. One-year CDS on investment-grade issuers trades in the tens of basis points and does not collapse toward zero as maturity shortens; short-dated spreads on stressed issuers can be enormous.

This is the same defect, in the same place, for the same reason as Section 21.7's. There, a diffusive volatility model could not produce the observed short-dated skew because continuous paths cannot generate short-horizon tails. Here, a diffusive asset process cannot produce short-dated credit spreads. Two literatures discovered the same limitation of continuous paths independently, and the repairs are related: jumps in both fields and, in credit, default made to arrive as a surprise rather than as the outcome of a diffusion. That second repair is the reduced-form approach.

24.8 REDUCED-FORM MODELS

Rather than explain default, model it directly. Let the default time τ be the first arrival of a Poisson process whose intensity λ_t is itself a stochastic process: a **Cox process**, or doubly stochastic Poisson process. Conditional on the whole path of the intensity, survival to T has the

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probability (24.2) already gave; averaging over intensity paths,

$$Q(T) = \mathbb{E}\left[\exp\left(-\int_0^T \lambda_u \, du\right)\right] \tag{24.7}$$

which is (24.2) with the expectation restored. Default is a surprise: however healthy the firm looks, the probability that it defaults in the next instant is $\lambda_t \, dt$, not zero. That is what the diffusion of Section 24.7 could not deliver, and it is why this construction repairs the short end: with default arriving as a jump, a one-month contract carries a spread of $\lambda(1 - R)$ from its first day.

Now price a bond with the same expectation as (24.7), because that step is where the framework shows its value. A zero-coupon bond paying 1 at T and nothing if the issuer has defaulted is worth

$$\bar{P}(0,T) = \mathbb{E}\left[e^{-\int_0^T r_u \, du} \, \mathbf{1}_{\{\tau > T\}}\right] = \mathbb{E}\left[\exp\left(-\int_0^T (r_u + \lambda_u) \, du\right)\right] \tag{24.8}$$

By (24.8) the hazard rate enters exactly where the short rate does, and that is the whole reason the approach took over. A defaultable bond is a riskless bond discounted at $r + \lambda$, so everything built for short-rate models in Chapter 31 (the trees, the closed forms, the calibration machinery) transfers to credit unchanged, with the intensity playing the part of a second rate factor. Nothing else in credit modeling is this convenient, and convenience is why the market prices with it.

Recovery slots into the same place. Suppose that on default the claim is worth a fraction R of what it was worth just before, so the loss is $1 - R$ of its *market value*, not of its face. Follow the bond over a short interval dt, writing $\bar{P}_t$ for its value at t. With probability $1 - \lambda_t \, dt$ the borrower survives and the bond is worth $\bar{P}_{t+dt}$; with probability $\lambda_t \, dt$ it defaults and the bond is worth $R \, \bar{P}_{t+dt}$. Under the risk-neutral measure the bond must earn the riskless rate in expectation, so

$$\begin{aligned} \bar{P}_t &= e^{-r_t \, dt}\left[(1 - \lambda_t \, dt) \, \bar{P}_{t+dt} + \lambda_t \, dt \, R \, \bar{P}_{t+dt}\right] \\ &= e^{-r_t \, dt}\big(1 - \lambda_t (1 - R) \, dt\big)\bar{P}_{t+dt} \approx e^{-(r_t + \lambda_t (1-R)) \, dt} \, \bar{P}_{t+dt}. \end{aligned}$$

Figure 24.4 draws the step. Default has added $\lambda_t(1 - R)$ to the one-period discount rate: the intensity of the event times the fraction it destroys. Chaining these steps from 0 to T and taking the expectation over the paths of r and λ gives

$$\bar{P}(0,T) = \mathbb{E}\left[\exp\left(-\int_0^T \big(r_u + \lambda_u(1 - R)\big) \, du\right)\right] \tag{24.9}$$

so the credit spread is $\lambda(1 - R)$. That is (24.3), Section 24.2's credit triangle, arriving as the constant-intensity case of a general result rather than as a rule of thumb. It also makes Example 24.3's identification problem structural rather than accidental: λ and R enter the price only through their product, so under this convention no bond price and no CDS spread can separate them.

One convention changed along the way. Section 24.2 took recovery as a fraction of *face*, which is how a CDS settles and how the market quotes R; the derivation above takes it as a fraction of the bond's *market value* an instant before default, which is what makes the algebra close. For a bond trading near par the two conventions agree, because face and market value then coincide. They differ for a bond far from par (a bond at 50 that recovers 40 has recovered 40% of its face but 80% of its market value), and for a CDS on such a bond the face convention is the one that matters, since the contract pays $1 - R$ of face whatever the bond was worth. Chapter 25 keeps the face convention throughout for that reason; the market-value convention

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is used here because it is the one under which the hazard rate enters a bond price exactly as a spread.

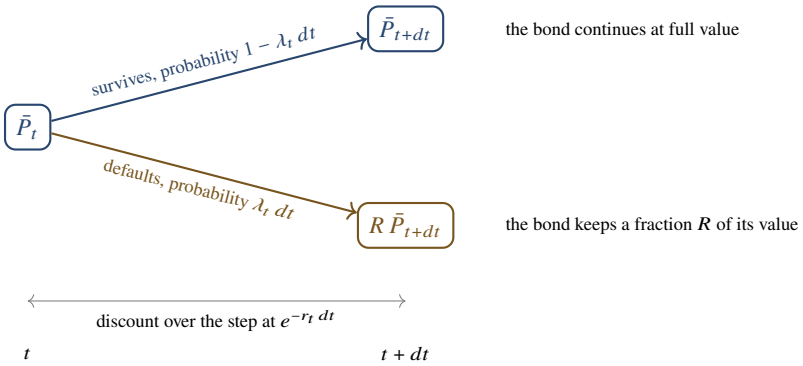


Figure 24.4: One step of the argument behind (24.9). The bond's value today is the discounted risk-neutral expectation of the two outcomes, which shortens to a discount rate of $r_t + \lambda_t(1 - R)$.

Each family has its strengths and its weaknesses.

Reduced-form models fit. Because λ is a free function, any observed spread curve can be reproduced exactly, which is Section 24.3's bootstrap, and it is the same property that makes local volatility the baseline in Chapter 21 for payoffs today's surface alone determines.

Structural models explain. They say why a spread is what it is, connect it to the equity market, and produce a hedge: if equity is a call on the firm, credit can be hedged with equity, and capital-structure arbitrage is that trade.

The two families also differ on correlation, where it matters most. In a structural model two firms default together because their asset values are correlated, a mechanism with an economic story attached. In a reduced-form model correlation must be imposed on the hazard rates, and the imposition is arbitrary. Chapter 26 is largely about what happened when the market chose a convenient imposition and treated it as a fact.

The market uses reduced-form for pricing and structural for thinking. A CDS desk bootstraps hazard rates; a credit analyst asks how far the firm is from default. The two are not in conflict, because they answer different questions.

24.9 REAL-WORLD AGAINST RISK-NEUTRAL

Section 9.2 computed CVA from the counterparty's CDS spread rather than from its historical default rate, and said that the default rate a spread implies has run several times the historical one. Here is how much.

Example 24.5 Most of a credit spread is not expected loss

The table sets five-year average cumulative default rates by rating for global corporate issuers, 1981–2024, from S&P Global Ratings' 2024 annual default study, against the risk-neutral probabilities implied by representative mid-cycle five-year spreads at 40% recovery. The spreads are illustrative; the historical rates are S&P's. AAA sits above AA only because few issuers were ever rated AAA, a small-sample artifact.

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Rating	Spread	Historical	Risk-neutral	Ratio
AAA	40 bp	0.34%	3.28%	9.6×
AA	55 bp	0.28%	4.48%	16.0×
A	80 bp	0.39%	6.45%	16.5×
BBB	130 bp	1.36%	10.27%	7.5×
BB	300 bp	5.75%	22.12%	3.8×
B	550 bp	15.60%	36.77%	2.4×

Decomposing the 130 bp BBB spread: expected loss at the historical default rate is the annual hazard $-\ln(1-0.0136)/5 = 0.274\%$ times $1-R$, or 16.4 bp, leaving 113.6 bp, so 87% of the spread is not expected loss.

The ratio of the two is largest for high-grade issuers. An AA or A issuer's risk-neutral default probability is about sixteen times its historical rate and a single-B issuer's under two and a half times; AAA is lower only because its small-sample historical rate exceeds AA's. Figure 24.5 draws the two columns on a log scale, where the vertical gap is the ratio, and the gap narrows steadily from A down to B. Whatever the excess compensates for, it does not scale with default risk.

Three things are in the residual, and separating them is a live question rather than a settled one. *A risk premium*, because defaults cluster in bad states and bearing them is unattractive for the reason Section 19.10 gave for volatility. *Illiquidity*, because corporate bonds trade rarely and the spread includes the cost of that. And *a premium for parameter uncertainty*, about the inputs themselves: the recovery of Example 24.3 is assumed, not known, and a seller of protection is being compensated for that too.

The practical consequence follows. Use risk-neutral probabilities to price and hedge, historical probabilities to forecast losses, and never mix them. An institution that provisions using risk-neutral probabilities will over-reserve by a factor of seven or more on investment-grade portfolios; one that prices using historical probabilities will systematically sell protection too cheaply and discover the difference when it tries to hedge.

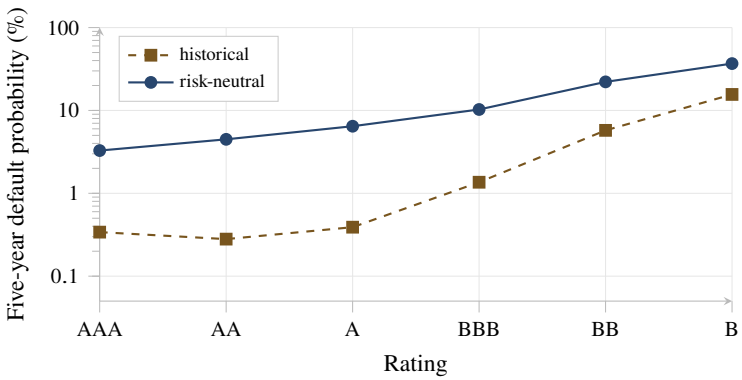


Figure 24.5: Five-year default probability by rating, historical and implied by spreads at 40% recovery, from Example 24.5. Log scale, so the vertical gap is the ratio.

24.10 RATINGS AND THEIR LIMITS

Ratings are an ordinal ranking of credit quality produced by agencies paid, for corporate and structured issues, by the issuer. Three properties determine how they should be used.

They rank well and calibrate poorly. The ordering has held up: over long samples, higher-rated issuers default less. But a corporate rating is defined as an ordinal opinion of

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relative credit risk, and no agency attaches a fixed probability to the letter. The mapping users work with is assembled after the fact from the agencies' own default studies (the source of Example 24.5), and it is much less stable than the ordering, drifting with the credit cycle and differing across asset classes. The one place an explicit letter-to-probability table did exist is structured finance. A **collateralized debt obligation** (CDO) is a pool of bonds or loans whose losses are divided into **tranches**, layers that absorb those losses in turn, and Moody's idealized default rates and the default-rate assumptions in S&P's CDO model were used to size the tranches. Chapter 26 is about what happened to that table: the same letter applied to a corporate bond and to a tranche described very different risks, because the tranche's rating depended on a correlation assumption that the letter did not disclose.

They are through-the-cycle by design. Agencies deliberately avoid re-rating on every market move, which makes ratings stable and makes them lag. An issuer whose spread has tripled may hold its rating for months. This is a feature for a bond mandate and a defect for a risk system, and confusing the two produces a risk report that says nothing has changed while the market says otherwise.

And they are used in regulation and contracts, which creates mechanical effects that have nothing to do with information. A downgrade below investment grade forces sales by mandate-constrained holders, triggers collateral calls under CSAs of the kind Chapter 8 described, and can raise funding costs discontinuously. A rating is therefore not merely an opinion about default; it is an event with contractual consequences, and the consequences can cause the deterioration the rating anticipated.

24.11 MIGRATION: HOW DEFAULT ACTUALLY HAPPENS

Firms rarely jump from health to default. They deteriorate, and the deterioration is what a rating records. Modeling that path requires a **transition matrix**: the probability of moving from each rating to each other rating over a year. Default is absorbing: an issuer that reaches it stays there.

Table 24.2: Illustrative one-year rating transition probabilities (%); rows are the starting rating, D is default.

	AAA	AA	A	BBB	BB	B	CCC	D
AAA	90.09	9.01	0.60	0.10	0.10	0.10	0.00	0.00
AA	0.60	90.59	7.91	0.60	0.10	0.20	0.00	0.00
A	0.05	2.10	91.40	5.50	0.60	0.20	0.10	0.05
BBB	0.02	0.20	4.20	89.90	4.30	0.90	0.20	0.28
BB	0.02	0.06	0.30	5.60	83.00	8.50	1.00	1.52
B	0.00	0.05	0.20	0.40	5.80	82.00	5.10	6.45
CCC	0.00	0.00	0.30	0.60	1.50	12.00	55.00	30.60

The matrix is illustrative, and from A down it is harsher than the long-run averages of Example 24.5: compounded to five years it gives default probabilities of 2.89% for BBB, 12.0% for BB and 31.9% for B, against S&P's 1.36%, 5.75% and 15.60%, so its pattern, not its levels, is what to carry over, and the two should not be mixed. Raising the matrix to the nth power gives n-year transitions, and the default column gives cumulative default probabilities. Horizons that are not whole years need one more step, which practitioners take and this chapter only names: find the **generator matrix** Λ with $P = e^{\Lambda}$, whose off-diagonal entries are the intensities of moving from one rating to another, extending the hazard rate of Section 24.1 from one exit to seven, and then $P(t) = e^{t\Lambda}$ gives the transition matrix for any horizon t. Not every annual matrix has a valid generator; Lando's book in the Further Reading treats the fitting.

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Example 24.6 Downgrade is the route to default

Table 24.2 places a BBB issuer after five years as follows:

	AAA	AA	A	BBB	BB	B	CCC	D
Probability (%)	0.10	1.37	14.64	62.17	12.90	5.06	0.87	2.89

So 78.3% is still investment grade, 18.8% has fallen to speculative grade (junk) without defaulting, and 2.89% has defaulted.

Now compare that 2.89% with the naive calculation. The one-year BBB default rate is 0.28%; compounding it as though each year were independent gives

$$1 - (1 - 0.0028)^5 = 1.39\%$$

against the matrix's 2.89%, a factor of 2.08.

The matrix gives more because default is reached through downgrade. A BBB issuer that is still alive in year four is disproportionately likely to be a BB or B issuer by then, facing a much higher hazard than the 0.28% it started with.

The route shows in the year-by-year numbers. Dividing each year's increase in the cumulative default probability by the probability of having survived to its start gives the **forward default probability** for that year, the discrete counterpart of the hazard rate of Section 24.1:

Year	1	2	3	4	5
Cumulative default probability (%)	0.28	0.72	1.31	2.03	2.89
Forward default probability (%)	0.28	0.44	0.59	0.74	0.87

The forward rate triples over five years although the starting rating has not changed, because the issuers still alive in year five are a different population from the ones that started: the ones that stayed BBB, plus the ones that fell to BB and B and have not yet defaulted. This is the same structure as the hazard curve of Example 24.2, arrived at from a different direction: the forward hazard rate of an investment-grade issuer rises with horizon, and here one can see the mechanism rather than merely the fitted curve.

Migration matters in two further ways. It is a risk even without default, because a downgrade widens spreads and triggers the contractual consequences of Section 24.10, so a portfolio marked to market bears it whether or not anything defaults. And the matrix is the natural input to a credit portfolio model; Section 24.14 builds one from it and measures that migration risk.

One caution follows: the matrix is not stable. Transition probabilities are strongly cyclical (downgrades cluster in recessions), so a matrix estimated over a long sample understates the risk of a bad year and overstates it in a good one, in the way Section 24.13's correlation is designed to capture.

24.12 SOVEREIGN CREDIT IS DIFFERENT IN KIND

A **sovereign** is a national government borrowing in its own name, and its debt comes in two kinds that the market names differently and this book has so far kept apart. **Local-currency debt** is issued in the government's own currency (the Treasuries of Chapters 4 to 6, and gilts; euro-area bonds such as Bunds are the borderline case, since the issuing government does not control the euro), and Part I treated it as the risk-free benchmark; it appeared later as the collateral everyone else posts. **External debt** (foreign-currency debt) is debt in a currency the government does not issue: dollars or euros borrowed by an emerging-market government from bondholders, banks, or official lenders. **Sovereign credit** is the risk that a government does not pay, and it lives mainly in the second kind, because a government can always print the currency it controls and cannot print anyone else's. CDS on emerging-market sovereigns reference that external debt; CDS on euro-area sovereigns, which do not control their currency, price the risk of redenomination as much as outright default. Everything above assumes a borrower that can

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be put through a bankruptcy process. A sovereign cannot, and the differences run deeper than the mechanics.

Ability and willingness are separate. A company defaults when it runs out of money. A country defaults when repaying costs more, politically, than not repaying—and it may have the money. Default is therefore a choice in a way that corporate default is not, which makes structural models of the kind in Section 24.5 inapplicable: there is no asset value crossing a debt threshold, because the relevant asset is the government's capacity to tax and its willingness to use it.

There is no court and no seniority. A defaulting sovereign cannot be liquidated, and the ranking of its obligations is whatever it chooses to respect. Restructurings are negotiated, protracted, and settled by a mixture of law and pressure. **Collective action clauses** (standard in New York-law sovereign bonds since Mexico's 2003 issue, and strengthened in 2014, after the Argentine holdout litigation, to a single vote aggregated across all of an issuer's series) bind dissenting holders to a majority agreement, and their presence or absence materially changes what a bond is worth in default.

The two kinds of debt are rated and priced apart. Default on local-currency debt is a policy choice with an inflationary alternative, so the market prices outright default and inflation together, as two routes to the same loss: a government that will not default outright may still repay in a currency worth less. Foreign-currency debt has no such escape, which is why the same issuer carries a local-currency and a foreign-currency rating, and why the second is usually the lower.

And the definition of default becomes contestable. A payment missed because the sovereign refuses is clearly a default; a payment missed because a third party blocked the transfer is less clear.

The pricing consequence is a wide and persistent basis between sovereign CDS and sovereign bond spreads, the difference Section 24.15 defines. The two should be close under the arbitrage that section describes, and they often are not, because the CDS references a definition of default while the bond references a cash flow, and because the natural sellers of sovereign protection are often institutions domiciled in the sovereign, which is Section 9.9's wrong-way risk in its purest form.

The point generalizes beyond sovereigns. Wherever default is a legal event rather than an economic one, the contract's definition matters as much as the borrower's finances, and Chapter 25 is largely about what happens when those two come apart.

24.13 DEFAULT CORRELATION

A portfolio of loans or bonds is not the sum of its parts, and the difference is entirely about correlation. State the problem first, because the device that solves it looks arbitrary until the problem is clear. Sections 24.2 to 24.4 deliver, for each borrower separately, the probability that it defaults by T; a portfolio's loss distribution needs the probability of every *combination* of defaults, and with n borrowers there are 2^n of them. Nothing in the data comes close to pinning that down: joint defaults are rare events, and for any particular group of borrowers a century of history holds few or none. So the joint distribution has to be built, and it has to be built under three constraints. It must reproduce each borrower's own default probability exactly, because the market quoted that and a dealer who disagrees with it has a position, not a model. It must carry dependence with a small number of parameters, because there is no data for more. And it must be computable for a portfolio of a hundred or more.

Correlating the default events themselves does not work. A default indicator is a zero-or-one variable, and the correlation of two such variables with different default probabilities is capped well below one. For probabilities $p_1 < p_2$ the ceiling is $\sqrt{p_1(1-p_2)/\big(p_2(1-p_1)\big)}$, which is

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0.43 at 2% and 10%: the rarer event sets it, because the two can default together no more often than the rarer one defaults at all. Nor can such correlations be chosen freely for more than two borrowers at once, since many matrices of pairwise indicator correlations correspond to no joint distribution at all. Merton's model of Section 24.5 suggests the way around: default happens when a continuous quantity, the firm's asset value, crosses a threshold, and continuous quantities are easy to correlate. So give each borrower a continuous latent variable standing in for its asset value, correlate the latent variables through the common state of the economy, and set each threshold so that the probability of crossing it equals that borrower's own market-implied default probability. The result is a Merton model calibrated to CDS curves instead of balance sheets, and it is what the credit market calls a **copula**: a function that supplies the dependence between variables whose marginal distributions are taken as given. Sklar showed in 1959 that any joint distribution can be split that way into its marginals and a copula, so the separation is always available; the choice made here is to take the copula from a multivariate normal.

The standard device is therefore the **one-factor Gaussian copula**. Give each borrower i a latent variable

$$X_i = \sqrt{\rho}\, M + \sqrt{1-\rho}\, Z_i, \qquad M, Z_1, \ldots, Z_n \text{ independent } N(0,1) \tag{24.10}$$

so that every X_i is standard normal and any two are correlated at ρ. Write $\pi_i(T) = 1 - Q_i(T)$ for borrower i's own probability of *defaulting* by T: the complement of the survival probability (24.2), and the marginal that Section 24.3's bootstrap produced from that borrower's CDS curve. Borrower i defaults when its latent variable falls below the threshold that marginal implies:

$$\text{default by } T \iff X_i < N^{-1}\big(\pi_i(T)\big) \tag{24.11}$$

The threshold is what makes this a copula. Each $\pi_i(T)$ is whatever the market said it was; the construction does not touch it. All (24.10) supplies is the *dependence* between the borrowers. Marginals from the market, dependence from an assumption: that separation is the entire appeal of a copula, and it is also what let a single number be asked to carry the joint behavior of the 125 issuers in a standard CDS index of Chapter 25.

The consequence is the one the section needs: conditional on the common factor, the borrowers are independent. In a homogeneous portfolio, where every borrower has the same default probability π by T, each defaults given M with probability

$$p(M) = N\left(\frac{N^{-1}(\pi) - \sqrt{\rho}\, M}{\sqrt{1-\rho}}\right) \tag{24.12}$$

so the number of defaults among n borrowers is a binomial averaged over the factor, weighted by the standard normal density ϕ of M:

$$\mathbb{P}(k \text{ defaults}) = \int_{-\infty}^{\infty} \binom{n}{k} p(M)^k \big(1 - p(M)\big)^{n-k}\, \phi(M)\, dM \tag{24.13}$$

One integral, and the whole portfolio is priced; Example 24.8 carries one case through it. That tractability, and not any evidence about how firms actually fail, is why this model became the market standard.

Two features of (24.12) recur in everything that follows. A bad common factor raises everyone's default probability at once: $p(M)$ rises as M falls, which is the clustering the model is built to produce. And the idiosyncratic channel diversifies away while the common one does not, so in a large portfolio the loss distribution is driven almost entirely by M.

The limit can be written down. Conditional on M the borrowers are independent with common default probability $p(M)$, so by the law of large numbers the fraction of a very large

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homogeneous portfolio that defaults is $p(M)$ itself:

$$\mathrm{DR}(M) = p(M), \tag{24.14}$$

a deterministic function of one standard normal draw, whose quantile Section 24.14 takes. Figure 24.6 shows what correlation does to the distribution of defaults, and Example 24.7 measures it.

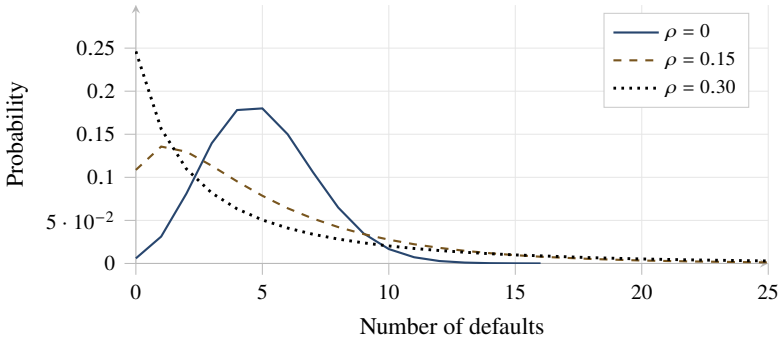


Figure 24.6: Distribution of defaults in a portfolio of 100 borrowers, each with 5% default probability. Expected defaults are 5 in every case.

Example 24.7 Correlation moves probability to both tails

Take one hundred borrowers, each with a 5% default probability over the horizon.

ρ	Expected	Std. dev.	P(no defaults)	P(20 or more)
0.00	5.00	2.18	0.59%	0.00%
0.05	5.00	3.22	3.13%	0.11%
0.15	5.00	4.89	10.86%	1.85%
0.30	5.00	7.12	24.62%	4.91%
0.60	5.00	11.56	53.02%	7.87%

The expected loss does not move at all. What correlation does is take probability out of the middle and put it at *both* ends, which is why the same expected loss can describe a safe portfolio or a dangerous one: at $\rho = 0.6$ there is a 53% chance that nothing defaults and an 8% chance that at least a fifth of the portfolio does.

That double-tailed behavior is what tranche pricing turns on. A senior tranche, one that suffers only if losses exceed some large threshold, is a bet on correlation and almost nothing else, because its expected loss is negligible at low correlation and material at high correlation while the portfolio's expected loss is unchanged. An equity tranche (the first-loss layer, which absorbs the portfolio's losses before any other tranche does) is the opposite: it benefits from correlation, because correlation raises the chance that nothing happens at all.

So a tranche is a correlation instrument that looks like a credit instrument, and pricing one requires a number, ρ, that no observable market quoted before tranches themselves began to trade. The circularity in that sentence is Chapter 26's subject.

Example 24.8 Applying the copula, one value of the common factor at a time

Here is how the $\rho = 0.15$ row of Example 24.7 is produced. With $\pi = 0.05$ the threshold in (24.11) is $N^{-1}(\pi) = -1.6449$, and conditional on the common factor M each of the 100 borrowers defaults

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independently with probability

$$p(M) = N\left(\frac{-1.6449 - \sqrt{0.15}\,M}{\sqrt{0.85}}\right),$$

which is (24.12). Given M, the number of defaults K is binomial with parameters 100 and $p(M)$, so every quantity of interest (the expected count $100\,p(M)$, the probability of no defaults $(1-p(M))^{100}$, the probability of twenty or more) is an elementary binomial calculation. The table evaluates them at six values of M, with the standard normal density $\phi(M)$ that weights each state.

M	$\phi(M)$	$p(M)$	$\mathbb{E}[K \mid M]$	$\mathbb{P}(K=0 \mid M)$	$\mathbb{P}(K \geq 20 \mid M)$
−3	0.0044	30.0%	30.0	0.00%	99.1%
−2	0.0540	17.3%	17.3	0.00%	27.1%
−1	0.2420	8.6%	8.6	0.01%	0.03%
0	0.3989	3.7%	3.7	2.26%	0.00%
1	0.2420	1.4%	1.4	25.03%	0.00%
2	0.0540	0.4%	0.4	64.72%	0.00%

The unconditional numbers are each column integrated against $\phi(M)$, which is (24.13) applied to the event in question: expected defaults 5.000, no defaults 10.86%, twenty or more 1.85%. These are the $\rho = 0.15$ row. Two things are visible in the table that the row alone hides. The expected count comes out at five because $p(M)$ is raised above 5% in the bad states and lowered below it in the good ones by amounts that average out exactly. And the two tails live in different states: three-quarters of the probability of twenty or more defaults comes from $M \leq -2$, states with about 2% of the probability, while two-thirds of the probability of none at all comes from $M \geq 1$. Raising ρ steepens $p(M)$ in M, feeding both tails at once from the middle, which is the redistribution Figure 24.6 draws.

24.14 CREDIT VAR

Chapter 38 asks how much a trading book could lose over a few days. A loan book needs the same question asked differently, for three reasons. Its positions cannot be sold in a few days, so the horizon is the time it takes for losses to be realized and capital to be rebuilt, conventionally one year. Its loss distribution is nothing like a trading book's roughly symmetric profit and loss: in most years a few borrowers default and the loss is close to its average, and in rare years many default together, so the distribution is bounded on one side and long on the other. And its expected loss is not a risk in the sense that matters for capital. It is a cost, charged in the loan's spread and set aside in provisions: the expected credit loss of IFRS 9, and the current expected credit loss (CECL) standard in the United States. What capital has to absorb is the loss in excess of that cost.

Credit VaR is therefore defined as a high quantile of the one-year portfolio loss minus the expected loss, a difference called the **unexpected loss**. The confidence level is far higher than Chapter 38's: 99.9% for regulatory capital, and typically 99.95% or more for a bank's own **economic capital**, a level chosen so that the bank's probability of being wiped out in a year matches the default rate of the rating it wants to hold.

The large-portfolio limit of Section 24.13 delivers this quantity directly. In a large homogeneous portfolio whose borrowers share a one-year default probability π, the fraction that defaults within the year is $\mathrm{DR}(M) = p(M)$ of (24.14), a decreasing function of the common factor, so its quantile at confidence q is $p(M)$ evaluated at $M = -N^{-1}(q)$. That quantile is the **worst-case default rate** (WCDR), the default rate exceeded in only a fraction $1-q$ of years:

$$\mathrm{WCDR}(q) = N\left(\frac{N^{-1}(\pi) + \sqrt{\rho}\,N^{-1}(q)}{\sqrt{1-\rho}}\right) \tag{24.15}$$

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With exposure at default (EAD) and recovery R, the credit VaR is

$$\text{Credit VaR}(q) = \text{EAD}\,(1 - R)\,\big[\,\text{WCDR}(q) - \pi\,\big] \tag{24.16}$$

The feature that matters most is visible at $\rho = 0$, where (24.15) returns π at every confidence level. A large portfolio of independent loans has no credit VaR at all, because the law of large numbers pins its default rate to the mean. Every dollar of a granular portfolio's unexpected loss comes from correlation, which Example 24.9 puts in numbers.

Example 24.9 All of the credit VaR is correlation

A bank holds a \$100 million book of corporate loans, numerous and small enough to be granular, each with a one-year default probability of 1% and a loss given default of 40%, the Basel foundation value for a senior unsecured loan to a nonfinancial corporate. The expected loss is $100 \times 0.40 \times 0.01 = \$0.40$ million whatever the correlation. The worst-case default rate (24.15) and the credit VaR (24.16) are not:

ρ	WCDR, 99%	WCDR, 99.9%	Credit VaR, 99%	Credit VaR, 99.9%
0.00	1.00%	1.00%	\$0.00m	\$0.00m
0.05	3.19%	4.67%	\$0.88m	\$1.47m
0.10	4.68%	7.75%	\$1.47m	\$2.70m
0.20	7.53%	14.55%	\$2.61m	\$5.42m
0.30	10.43%	22.44%	\$3.77m	\$8.58m

Take the $\rho = 0.20$ row, close to the correlation the Basel formula assigns to a borrower with this default probability (Section 39.10). In the worst year in a thousand, 14.55% of the book defaults, against 1% on average, and the loss is $100 \times 0.40 \times 0.1455 = \$5.82$ million. Subtracting the \$0.40 million already priced leaves a credit VaR of \$5.42 million, 13.6 times the expected loss.

Two consequences follow. Correlation sets the ratio of credit VaR to expected loss, so two books with identical expected loss can need capital that differs by a factor of almost six (\$1.47 million at $\rho = 0.05$ against \$8.58 million at 0.30). And the confidence level matters as much as the model: moving from 99% to 99.9% raises the credit VaR to between 1.67 and 2.3 times its 99% value, more at higher correlation, because the tail is long and correlation is what lengthens it.

Two assumptions sit inside (24.15), and each understates the risk of a real book. The portfolio is taken to be infinitely granular. A book with a few large exposures keeps idiosyncratic risk that does not diversify, and its credit VaR is higher than the formula says, by an amount that grows with the concentration. And the dependence is the Gaussian copula, with the absence of tail dependence that Chapter 26 shows to matter and Section 38.12 treats formally; a copula with fatter joint tails puts more defaults in the bad years and raises the quantile further.

Migration, not only default. The measure above counts a loss only when a borrower defaults, which is right for a loan held at amortized cost. A portfolio marked to market loses money whenever a borrower is downgraded and its spread widens, which is the migration risk of Section 24.11. **CreditMetrics**, published by J.P. Morgan in 1997, measures it. It revalues each position at the horizon in every rating state, discounting the remaining cash flows on that rating's spread curve, and weights the values by a row of the transition matrix. To correlate migrations across borrowers it reuses the latent variable of (24.10), cutting that variable into bands rather than at a single default threshold: a BBB borrower defaults if X_i falls below $N^{-1}(0.28\%)$, ends the year at CCC or worse if it falls below $N^{-1}(0.48\%)$, and so on up the row. The same ρ then correlates downgrades as well as defaults, and a portfolio's distribution is found by simulating M and the Z_i.

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Example 24.10 Downgrade, not default, sets the 99% VaR of a BBB bond

A five-year BBB bond pays an annual coupon of 5% on \$100 face. A year from now it will have paid one coupon and have four years left, and its value then depends on its rating. The risk-free rate is 3% for every maturity, compounded annually, each rating trades at the spread in the table, the spreads of Example 24.5 extended to CCC, and the bond recovers \$40 if it defaults during the year, without the coupon. The probabilities are the BBB row of Table 24.2.

Rating in one year	Probability	Spread	Value
AAA	0.02%	40 bp	110.89
AA	0.20%	55 bp	110.32
A	4.20%	80 bp	109.38
BBB	89.90%	130 bp	107.52
BB	4.30%	300 bp	101.53
B	0.90%	550 bp	93.54
CCC	0.20%	1,200 bp	76.45
Default	0.28%	—	40.00

Each value is the coupon just received plus the remaining flows discounted at 3% plus the spread; at BBB it is $5 + 5/1.043 + 5/1.043^2 + 5/1.043^3 + 105/1.043^4 = 107.52$. The probability-weighted mean is 106.97.

Reading up from the bottom of the table, the bond ends the year in default with probability 0.28%, at CCC or worse with 0.48%, and at B or worse with 1.38%. The first percentile therefore falls in the B state, and the 99% credit VaR is $106.973 - 93.535 = 13.44$ per \$100 of face. The 0.1% quantile falls in default, and the 99.9% credit VaR is $106.973 - 40.00 = 66.97$.

Now drop migration and keep only default, so the bond is worth 107.52 unless it defaults. Its mean is 107.33, and since default has a probability below 1% the first percentile is 107.52 itself, above the mean: the default-only model reports no 99% credit VaR at all, where the migration model reports 13.44. At 99.9% the two agree to within half a dollar, because both quantiles sit in default.

So the moderate quantile of a high-grade bond is a question about downgrade and the extreme quantile is a question about default. That is why the Basel formula of Section 39.10 carries an adjustment for maturity, which charges a long loan for the downgrade risk it carries before it can default. It is also why the trading book's default charge of Section 39.3 leaves migration to the spread sensitivities, which already measure what a downgrade costs.

Who uses it. A bank's credit portfolio management function runs a model of this kind across the whole loan book to set concentration limits and to allocate economic capital loan by loan, and the allocation enters pricing: under **risk-adjusted return on capital** (RAROC), a loan must earn its hurdle return on the capital it consumes, so a loan to a borrower highly correlated with the rest of the book carries a wider spread than one that diversifies it. The allocation is a contribution to the portfolio quantile, the same decomposition Section 38.15 applies to market risk. Where the model finds a concentration, the bank buys CDS protection on it (Chapter 25) or transfers a tranche of the book in a synthetic securitization (Chapter 26), and Section 27.11 shows how much regulatory capital each of those relieves. Rating agencies used the same machinery on the pools behind structured credit. And supervisors turned (24.15) into the formula that sets most banks' credit capital, with a prescribed correlation and a maturity adjustment. They could do so because, as Gordy (2003) showed, in a one-factor model of a granular portfolio the portfolio's quantile is the sum of each loan's own worst-case loss, so (24.15) can be applied one loan at a time with each loan's own π. Section 39.10 takes the formula up beside the output floor that now limits it.

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24.15 THE CDS–BOND BASIS

Two instruments reference the same issuer: a bond, whose spread over the risk-free curve compensates for default risk, and a CDS, whose spread does the same. They should agree, and often they do not.

The bond's spread needs defining first, because there are several. The one that compares with a CDS is the **Z-spread**: the constant spread that, added to every rate on the risk-free curve, discounts the bond's cash flows to its market price. The curve is the SOFR swap curve, which is what a CDS dealer funds and discounts on; a spread over Treasuries mixes in the swap spread of Chapter 7. A close cousin is the par **asset-swap spread**, the spread over SOFR received when the bond's fixed coupons are swapped to floating, which is how a basis trade is packaged in practice and is close to the Z-spread for a bond near par. From the Z-spread, a hazard rate follows the same two ways it did for a CDS: roughly through the credit triangle, and exactly by pricing the bond.

Example 24.11 A hazard rate from a bond price

A five-year bond pays an annual coupon of 5% and trades at 98.55. The SOFR curve is flat at 3%, continuously compounded. Discounted on that curve alone the cash flows would be worth

$$5\,(e^{-0.03} + e^{-0.06} + e^{-0.09} + e^{-0.12}) + 105\,e^{-0.15} = 108.94,$$

and the bond's Z-spread, the z that makes the same sum at $3\% + z$ equal 98.55, is 220 bp.

The credit triangle reads the spread as $\lambda(1 - R)$, so at 40% recovery $\lambda = 0.022/0.6 = 3.67\%$.

Pricing the bond with a constant hazard rate, recovery of 40% of face paid at default, and coupons only while the issuer survives:

$$98.55 = \sum_{t=1}^{5} 5\,e^{-(0.03+\lambda)t} + 100\,e^{-(0.03+\lambda)5} + 40\,\lambda \int_0^5 e^{-(0.03+\lambda)t}\,dt$$

gives $\lambda = 3.63\%$, close to the triangle, as it should be for a bond near par. The gap of $108.94 - 98.55 = 10.39$ between the riskless and the actual price is the present value of the expected loss from default, and the equation above is that statement written out.

The same issuer's five-year CDS trades at 170 bp, which implies $\lambda = 2.83\%$. The bond and the CDS disagree about the issuer's hazard rate by about 0.8 percentage points, and Example 24.12 is about why that disagreement can persist.

The arbitrage argument appears complete. Buy the bond, buy protection on the issuer, and the combination has no credit risk: if the issuer defaults, the protection makes up the loss. So the bond's spread should equal the CDS spread, and the difference, the **basis**, should be zero.

$$\text{basis} = s_{\text{CDS}} - s_{\text{bond}} \tag{24.17}$$

Example 24.12 shows why a nonzero basis can persist.

Example 24.12 Why a negative basis is not an arbitrage

The bond of Example 24.11 trades at a Z-spread of 220 bp; the five-year CDS trades at 170 bp. The basis (24.17) is −50 bp. The trade suggests itself: buy the bond, buy protection, and collect $220 - 170 = 50$ bp on a package with no credit exposure.

Except that the bond must be financed. Buying it requires cash, raised in repo against the bond itself. A corporate bond does not repo at the risk-free rate: with a haircut and a funding spread the all-in cost might be 60 bp over the curve the 220 bp was measured against, and $50 - 60 = -10$ bp. The trade loses money. And it consumes balance sheet, which under the frameworks of Chapter 27 carries a capital charge of its own.

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So the negative basis is not an arbitrage but a measure of what balance sheet costs, and it persists while funding costs more than the basis pays. Several other factors move the basis, each in a known direction.

The cheapest-to-deliver option pushes the basis up. A CDS settles at the recovery price an auction sets for the cheapest qualifying obligations, so the protection buyer is in effect delivering the cheapest bond. That right raises the CDS spread by its value; it is the structure of the Treasury futures delivery option of Chapter 6.

Counterparty risk pushes it down. Protection is worth less than face if the seller might fail, and least when seller and reference entity are likely to fail together, which is Section 9.9's wrong-way risk.

And the contracts define default differently. A CDS pays on credit events that may include restructuring, a broader trigger than the missed payment that defaults a bond, so protection is worth more and the basis rises (Section 25.10).

At the trough of the 2008 crisis the basis on investment-grade issuers reached about −240 bp on average, and about −560 bp on speculative-grade issuers (Bai and Collin-Dufresne 2019), an enormous riskless profit on the arbitrage logic. It was not taken, because those who would have taken it could not fund the bond at any price, the limit of arbitrage of Section 5.15.

24.16 WHO BEARS CREDIT RISK

Banks originate it and, since the capital reforms of Sections 27.11 and 26.13, increasingly distribute it; what they retain is capitalized under the internal-ratings formula of Section 39.10, the regulatory form of credit VaR, and the charge often decides whether a loan is made. **Bond investors** hold it deliberately, paid by the spread of Example 24.5, mostly not for expected loss. **Insurers and pension funds** are the natural long-horizon holders, because they can collect the spread's illiquidity premium without being forced to sell. **Protection sellers in the CDS market** take it synthetically, through Chapter 25's contract.

Private credit funds have taken a growing share of what banks once held. The shift has two sides: the risk sits largely with locked-up long-horizon capital, an improvement (though retail semi-liquid vehicles rationed redemptions in 2026), but it is valued infrequently and by the holder, so its price is far harder to observe. Chapter 26 records what happened the last time credit risk migrated to holders whose marks no one could check.

24.17 WHERE THIS CAME FROM

Merton (1974) began the subject: equity is a call on the firm's assets, and debt a short put. Black and Cox (1976) added the covenants that let default happen before maturity, and Leland (1994) made the capital structure itself endogenous. The structural approach is illuminating but inaccurate: it predicts short-maturity spreads near zero. The reduced-form answer modeled the arrival of default instead: Jarrow and Turnbull (1995) and Duffie and Singleton (1999) founded it, Lando (1998) supplied the Cox-process formulation, and Hull and White (2000) gave Chapter 25's CDS valuation.

Altman (1968) had earlier combined accounting ratios in a discriminant score to predict bankruptcy. On the portfolio side, KMV commercialized distance to default in the 1990s; J.P. Morgan published CreditMetrics in 1997; Li (2000) put a Gaussian copula on default times; and Vasicek's large-portfolio limit became Basel II's risk-weight formula.

SUMMARY

Credit risk is two questions, default probability and recovery, and market prices identify only their product. The hazard rate is the instantaneous conditional default rate, so survival is $\exp\left(-\int_0^T \lambda(u)\,du\right)$ and default probabilities compound rather than add.

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The credit triangle $s = \lambda(1 - R)$ is exact for a continuously paid premium, because survival and discounting cancel between the legs. Hazard and recovery are not separately identified: the same CDS spread is reproduced by a low recovery with a low hazard or a high recovery with a high hazard, which is harmless for anything that depends only on the product and decisive for a digital default swap or a regulatory calculation that asks for the two separately. Recovery itself ranks with priority of claim and is lowest when defaults are highest, so a fixed recovery convention understates tail losses twice.

In Merton's model equity is a call on the firm's assets struck at the face value of debt, so lending is selling a put on those assets, and equity holders gain from the volatility that debt holders lose from. The model's spread goes to zero at short maturity, for the reason Section 21.7 found diffusive volatility models unable to make a short-dated skew. Reduced-form models repair it by making default a surprise with a free hazard function, which fits any curve and explains nothing.

Risk-neutral default probabilities exceed historical ones, and most of an investment-grade spread is not expected loss: price with risk-neutral probabilities, forecast losses with historical ones, and never mix them. Ratings rank well, calibrate poorly, and lag by design. Default is reached through downgrade, so a rating transition matrix says more than compounding the one-year rate, and for a high-grade bond the moderate quantile is a question about downgrade and the extreme one a question about default. A sovereign's default is partly a choice, so structural models do not apply and the contract's definition of default matters as much as the borrower's finances.

Correlation leaves a portfolio's expected loss unchanged and moves probability to both tails, which is why a senior tranche is a bet on correlation and why Chapter 26 is about a number no market quoted. The same mechanism sets credit VaR, the loan book's version of Chapter 38's question: a one-year horizon, a confidence level of 99.9% or more, and only the loss beyond the expected loss, which is priced and provisioned rather than held as capital. With independent defaults a large portfolio's worst-case default rate equals the average, so all of a granular book's credit VaR comes from correlation. The CDS–bond basis persists because closing it takes funded balance sheet, and it widens when that is scarce.

FURTHER READING

Merton (1974) is short and worth reading in the original: Merton states his assumptions plainly, and the model's limitations follow directly from them. Crosbie and Bohn (2003) document the KMV implementation of Section 24.6, including the default point and the mapping to expected default frequencies. Duffie and Singleton's *Credit Risk* (2003) is the standard treatment of reduced-form models and takes up the identification problem of Example 24.3; Lando's *Credit Risk Modeling* (2004) is the other standard, and the one to consult for rating-transition models and the generator matrix that Section 24.11 names. Vasicek's (2002) short note is the source of the large-portfolio limit (24.14). Gupton, Finger and Bhatia's *CreditMetrics—Technical Document* (1997) is the original statement of the migration approach of Section 24.14, and Gordy (2003) is the paper that shows why the large-portfolio quantile can be computed one loan at a time, the property the Basel formula rests on. On recovery, the empirical literature is larger than the 40% convention suggests, and less reassuring; the positive correlation between default rates and loss given default is the finding that recurs across samples. S&P Global Ratings' annual default study is the source of the historical figures in Example 24.5 and Moody's of Table 24.1; both are worth consulting directly, for the data and for the methodological notes that explain why the numbers move between vintages. On the decomposition of credit spreads, the "credit spread puzzle" literature is large and unresolved. Li (2000) is the paper that put the copula of Section 24.13 into the market, and is worth reading beside Chapter 26's account of

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what followed.

PRACTICE QUESTIONS

24.1 A borrower has a constant hazard rate of 4%. Compute the probability of default within five years and within ten, and explain why a 3% hazard rate does not mean a 30% probability over ten years.

24.2 A CDS trades at 240 bp with assumed recovery of 40%. Estimate the hazard rate, and explain why a change in the discount curve alone barely moves a par CDS spread.

24.3 Explain why the credit triangle is exact rather than approximate, and what the usually quoted error actually measures.

24.4 Using Example 24.2, explain why applying the credit triangle to a five-year spread understates the hazard rate over years three to five.

24.5 A five-year CDS trades at 170 bp. State the implied hazard rate at 40% recovery and at 60%, and name one instrument the two assumptions would price differently.

24.6 Explain why recoveries are lowest in the years when defaults are highest, and what this does to portfolio losses.

24.7 Explain why equity is a call option on the firm and debt is short a put, what this implies about the shareholder–creditor conflict, and what the ratio of equity to asset volatility measures as the firm approaches default.

24.8 Explain why Merton's credit spread goes to zero as maturity shortens, and name the chapter in which the same defect appeared in a different market.

24.9 Historical five-year default rates for BBB issuers are about 1.36% and the spread is about 130 bp. Compute the expected loss, state what fraction of the spread it accounts for, and explain why the ratio of risk-neutral to historical default probability is much larger for investment-grade than for single-B issuers.

24.10 A portfolio of 100 borrowers has 5% default probability per borrower. Explain why raising correlation leaves expected defaults unchanged but raises both the probability of zero defaults and the probability of twenty or more, and why a senior tranche is a bet on correlation while an equity tranche benefits from it.

24.11 Using Table 24.2, state the probability that a BBB issuer is still investment grade after five years and the probability it has fallen to speculative grade without defaulting, then explain why compounding a one-year default rate understates the five-year probability and give the factor from Example 24.6.

24.12 State two reasons a structural model cannot be applied to a sovereign borrower, and explain why the natural sellers of sovereign protection create wrong-way risk.

24.13 Explain why fixing recovery at 40% is harmless for pricing a CDS and damaging for a capital calculation.

24.14 A large, granular loan book has a one-year default probability of 2%, a loss given default of 45% and a copula correlation of 0.15. Compute the worst-case default rate at 99.9% and the credit VaR per \$100 million, and explain why the credit VaR would be zero if the loans defaulted independently.

24.15 Using Example 24.10, explain why the 99% credit VaR of a BBB bond is set by downgrade and its 99.9% credit VaR by default, and state what a model that ignores migration reports at each level.

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24.16 Explain why credit VaR subtracts the expected loss while Chapter 38's VaR is measured from zero, and why its horizon is a year rather than a few days.

24.17 A five-year bond with a 5% annual coupon trades at 97.00 against the flat 3% SOFR curve of Example 24.11. Find its Z-spread and the hazard rate it implies at 40% recovery by the credit triangle, and state the CDS–bond basis if the issuer's five-year CDS trades at 170 bp. Explain why a basis trader would not simply buy the bond and buy protection.

FURTHER QUESTIONS

24.18 Example 24.1 shows the credit triangle is exact for a continuously paid premium and 0.376% off for a quarterly contract with accrual. Derive the cancellation, identify precisely which contract features are required for exactness, and determine how the relation changes for a contract that pays the premium annually in arrears with no accrual on default.

24.19 Example 24.3 shows that a CDS spread does not identify λ and R separately. Construct a pair of instruments whose prices together *would* identify them, explain why such instruments are illiquid or absent in practice, and discuss what this implies for the regulatory requirement to report probability of default and loss given default separately.

24.20 Section 24.7 argues that Merton's short-maturity failure is the same defect Section 21.7 found in diffusive volatility models. Make the parallel precise: state the shared mathematical property, show why it forces both spreads and skews to vanish at short horizons, and evaluate whether the repairs adopted in the two literatures are the same repair.

24.21 Example 24.5 finds that 87% of a BBB spread is not expected loss and that the ratio falls steeply from the high-grade ratings to single-B. Propose a decomposition of the residual into risk premium, illiquidity, and parameter uncertainty; state what data would identify each component; and explain why the rating-dependence of the ratio is evidence about which component dominates.

24.22 Section 24.5 states that lending is selling a put on the borrower's assets. Use this to derive the effect on debt value of an increase in asset volatility, a dividend payment to shareholders, and a debt-financed acquisition. Then explain which covenants in a real loan agreement correspond to which of these, and why covenants are the creditor's substitute for a hedge they cannot execute.

24.23 Example 24.7 shows correlation moving probability to both tails. Derive the limiting loss distribution as $\rho \to 1$, explain what it says about the meaning of a AAA rating on a senior tranche, and use this to state precisely what a rating agency was asserting when it rated such a tranche.

24.24 Section 24.16 argues that the migration of credit from banks to private funds genuinely has two sides. Set out the argument that it has made the system safer and the argument that it has made risk unobservable, identify what evidence would distinguish them, and explain why the absence of a market price is not the same as the absence of a loss.

24.25 Section 24.14's worst-case default rate assigns each loan a capital charge that depends only on that loan. Show that in the one-factor large-portfolio model the portfolio's loss quantile is the sum of the loans' own worst-case losses, explain why a regulator writing a capital rule needs this property, and identify two features of a real book that break it (a concentration and a second systematic factor), stating the direction in which each moves the true quantile.

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